

A Global Bang–Bang Value Formula for the Bounded Double Integrator

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Abstract

For the physical double integrator $\dot{x} = v$, $\dot{v} = u$, $|u| \leq a$, we close the minimum-time problem to rest at the origin on the entire state plane. The switching function $F_a = x + v|v|/(2a)$ selects either direct braking or a unique one-switch control. We give the two arc lengths, switch state and a closed value function, and verify exact termination and the Hamilton–Jacobi equation off the nonsmooth switching curve. A sharp rearrangement bound for the endpoint control moments supplies a global sufficiency proof; the affine Pontryagin switching function is an independent structural check. Origin, zero acceleration, reflection and parabolic scaling are retained explicitly. Executable exact/high-precision, symbolic, replay and hostile-mutation audits accompany the theorem.

1 Convention and synthesis

Fix $a > 0$ and let $T_a(x, v)$ be the least physical time for

$$\dot{x} = v, \quad \dot{v} = u, \quad |u| \leq a$$

to reach $(0, 0)$. Set

$$F_a(x, v) = x + \frac{v|v|}{2a}.$$

This convention fixes the state ordering, acceleration sign, terminal state and time clock; no rescaled or fitted clock is used.

Theorem 1 (Whole-plane minimum-time synthesis). *At $(0, 0)$, $T_a = 0$. If $F_a = 0$ and $v \neq 0$, the direct brake $u = -a \operatorname{sign} v$ is optimal and $T_a = |v|/a$.*

If $F_a \neq 0$, let $s = \operatorname{sign} F_a$ and

$$D = \frac{v^2}{2a^2} + \frac{sx}{a}.$$

Then $D > 0$ and

$$t_1 = \frac{sv}{a} + \sqrt{D} \geq 0, \quad t_2 = \sqrt{D} > 0, \quad T_a(x, v) = \frac{sv}{a} + 2\sqrt{D}. \quad (1)$$

Up to null sets, the optimal control is $u = -sa$ for time t_1 and $u = sa$ for time t_2 . Its switch state is

$$(x_1, v_1) = \left(\frac{saD}{2}, -sa\sqrt{D} \right), \quad (2)$$

and the second arc terminates at $(0, 0)$. Moreover,

$$T_a(-x, -v) = T_a(x, v), \quad T_a(\lambda^2 x, \lambda v) = \lambda T_a(x, v) \quad (\lambda > 0). \quad (3)$$

Algebraic construction. The assertion $D > 0$ follows directly from $sF_a > 0$; if $sv < 0$, that same strict inequality gives $\sqrt{D} > |v|/a$, and otherwise $t_1 \geq 0$ is immediate. Integrating the first constant-control arc gives (2). Since $v_1|v_1| = -sa^2D$, (2) lies on $F_a = 0$. The second arc has duration $\sqrt{D}$, so $v_1 + sa\sqrt{D} = 0$ and $x_1 + v_1\sqrt{D} + saD/2 = 0$. Reflection and scaling follow by substitution in (1). The global lower bound is proved below. $\square$

2 Two independent optimality certificates

For a normal minimum-time extremal, the costate satisfies $\dot{p}_x = 0$ and $\dot{p}_v = -p_x$. Hence p_v is affine in time and the minimizing control $u = -a \operatorname{sign} p_v$ switches at most once. This explains the structure but, as a necessary condition, is not by itself the global sufficiency proof.

Off $F_a = 0$, differentiating (1) gives

$$(T_a)_x = \frac{s}{a\sqrt{D}}, \quad (T_a)_v = \frac{s}{a} + \frac{v}{a^2\sqrt{D}}.$$

The inequality used above gives $\operatorname{sign}(T_a)_v = s$, and therefore

$$1 + v(T_a)_x - a|(T_a)_v| = 0. \quad (4)$$

Thus the displayed control attains the minimum in the HJB equation wherever the value is smooth.

Source note

Romano and Curti [1] treat minimum-time bounded normal LTI systems, reduction to an origin transfer, and the double-integrator context. We claim no priority for bang–bang synthesis or Pontryagin’s principle; the release contribution is the convention-locked theorem/boundary synthesis and reproducible audit.

References

- [1] M. Romano and F. Curti, *Time-optimal control of linear time invariant systems between two arbitrary states*, *Automatica* 120 (2020), 109151. DOI: 10.1016/j.automatica.2020.109151.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero or operator claim. **Data and code.** The theorem, canonical ledger and independent audits are released with HCS-C222. **AI-use disclosure.** Generative tools assisted drafting and code generation; the internal artifact chain checked the displayed claims and metadata. This is not external peer review.